

Part C — Reflection Paper

Wholesale Customer Segmentation Analysis

Course: Machine Learning — Fundamentals and Practice

Topic: Unsupervised Learning: Wholesale Customer Segmentation

1. Project Implementation Overview

In this practical assignment, I successfully built and evaluated a complete unsupervised learning pipeline designed to perform customer segmentation for a wholesale distributor. The primary goal was to uncover hidden purchasing behaviors within an unlabeled commercial dataset (`raw_wholesale_customers.csv`).

The implementation carefully followed the standard preprocessing workflow by isolating the six core continuous spending metrics — Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicassen. Non-spending categorical metadata columns (Channel and Region) were intentionally excluded during training to avoid distorting the multidimensional distance metrics.

To clean the data, I calculated the Interquartile Range (IQR) for each product category and applied a $k = 1.5$ bound multiplier to `.clip()` extreme spending amounts. This step was crucial for managing outliers without shrinking the dataset. The adjusted continuous values were then standardized using `StandardScaler` to bring all product categories to a uniform scale.

I then ran an Elbow Method loop from $k = 1$ to 10 to track the Sum of Squared Errors (SSE / Inertia). Based on the results, I trained a baseline K-Means clustering model ($K = 5$) and an independently researched second algorithm, Agglomerative Hierarchical Clustering, comparing both using Silhouette analysis and cluster centroid tracking.

2. Segment Interpretation & Business Action Strategies

By applying an inverse transformation to the final K-Means cluster centers, I mapped the scaled coordinates back to their original currency spending values. This allowed me to clearly interpret the purchasing profiles of the different customer groups.

Cluster 1 — High Fresh + Low Grocery / Detergents

This segment represents clients whose purchases are heavily weighted toward fresh produce, averaging over \$8,300 in Fresh goods while showing minimal spend on processed shelf items. In plain terms, these are typically traditional restaurants, fresh food cafés, or local greengrocers.

Suggested Business Action: The distributor should offer these clients volume-based discounts on seasonal produce or pitch a fresh food subscription service to capture consistent weekly orders.

Cluster 4 — High Grocery + High Detergents & Paper

This group shows a very different profile, spending heavily on non-perishable goods (averaging over \$18,300 in Grocery and \$7,700 in Detergents_Paper). These patterns point toward institutional buyers such as supermarkets, corporate supply chains, or big-box retail stores.

Suggested Business Action: The distributor can introduce bulk supply agreements for non-perishable essentials and cross-sell commercial cleaning packages to maximize the value of this high-spend segment.

Cluster 2 — The VIP All-Round Bulk Buyers

This exclusive segment shows exceptionally high spending across all categories, including large orders of Fresh (\$17,461), Milk (\$13,805), and Grocery (\$17,524). These are premium, high-volume clients, such as boutique hotel networks or regional distribution hubs.

Suggested Business Action: Assign a dedicated account manager to these clients and offer premium supply chain perks, such as priority delivery windows, to protect and grow these highly profitable relationships.

3. Understanding K-Means Clustering

In my own words, K-Means is a partition-based clustering algorithm that groups unlabeled data by organizing points around centrally calculated hubs called centroids. The variable K represents the number of distinct groups the user instructs the algorithm to find before processing begins.

The algorithm establishes its structure through an iterative assign-and-update loop. During initialization, the algorithm places K random coordinate markers, or centroids, into the multi-dimensional feature space. In the assignment step, it calculates the straight-line (Euclidean) distance from every data point to all K centroids and assigns each point to the cluster of its nearest centroid. In the update step, once all points are grouped, the algorithm recalculates each centroid's position as the mathematical mean of all data points currently assigned to that group.

This loop repeats: centroids shift position, and data points change cluster assignments accordingly. The process stops when the centroids settle into stable locations and points no longer switch groups, indicating that the model has successfully minimized the internal variance within each cluster.

4. Researched Second Algorithm: Agglomerative Clustering

For the alternative clustering method, I selected Agglomerative Hierarchical Clustering.

Why It Fits Wholesale Segmentation

Wholesale distribution data often contains naturally nested customer structures. For example, local cafés buy similarly to fine-dining restaurants, and both link into broader hospitality profiles. Agglomerative clustering captures these relationships by building a bottom-up hierarchical tree, or dendrogram. This approach avoids K-Means' rigid assumption that all clusters must be roughly spherical and equal in size, offering a more flexible view of purchasing behavior.

Core Insights Learned

How it Works: The algorithm begins by treating every customer as an independent, single-member cluster. It then scans the entire feature space, identifies the two closest points, and links them into a pair. This merging process repeats step-by-step, combining the closest clusters until all data points are joined into a single master tree.

Main Advantage: It does not require choosing a set number of clusters in advance and provides an intuitive, visual view of how different customer tiers link together.

Main Limitation: It is computationally demanding — calculating distances across large datasets requires significant memory and processing time, making it less scalable for high-volume customer databases.

Silhouette Score Comparison

Evaluating the two models against the preprocessed features revealed a clear performance difference: the K-Means model achieved a Silhouette Score of 0.2831, while Agglomerative Clustering achieved 0.2185. The higher silhouette score confirms that K-Means generated cleaner customer partitions with less overlap between groups, making it the more effective option for this particular feature layout.

5. Final Findings and Operational Recommendation

Based on the experimental metrics, structural outcomes, and practical business requirements, I recommend using the K-Means clustering approach ($K = 5$) for this wholesale customer segmentation task.

From a mathematical perspective, K-Means achieved a significantly higher Silhouette Score (0.2831 versus 0.2185), proving that its centroid-based approach creates tighter, more distinct customer groups. The cluster centers generated by K-Means also mapped cleanly to realistic, actionable business profiles, such as fresh food restaurants versus high-volume retail buyers.

Operationally, K-Means is highly efficient and easily accommodates new data points. As the distributor signs new clients, their profiles can be instantly assigned to the existing centroids without needing to recalculate the entire customer base. This clear separation and operational scalability make K-Means the ideal choice for driving the distributor's personalized marketing and inventory strategies.